INDIAN FOOD ML PROJECT — COLLEGE VIVA & DEFENSE GUIDE

Comprehensive Question-Answer Guide & Key Metrics Cheat Sheet

1. KEY METRICS CHEAT SHEET (MEMORIZE THESE NUMBERS!)

Metric Description	Value	Notes / Model Details
Unified Dataset Records	8,411 clean rows	Ingested across 3 sources (USDA, OFF, Indian Data)
Indian Food Dishes	1,014 dishes	Includes regional (North/South/West/East) & diet tags
Calculated Composite Meals	200 recipes	Computed using weighted ingredient sum formula
Best Protein Regressor	MAE = 1.33g, R^2 = 0.6659	Random Forest Regressor (Expanded Dataset)
Best Quality Classifier	96.77% Accuracy	Gradient Boosting Classifier (Macro F1 = 0.9239)
Recommendation Engine	Vector Cosine Sim	StandardScaler + Hard Numerical Constraints + Fallback

2. THE 1-MINUTE OPENING ELEVATOR PITCH

Say this when the examiner asks 'Explain your project':
*'Respected Examiners, my project is a **Multi-Source Machine Learning System for Food Nutrient Prediction, Nutritional Quality Classification, and Nutrition-Aware Food & Meal Recommendation**, with a special focus on Indian Food Composition.*

*We unified **8,411 food records** across USDA, Open Food Facts, and Indian composition databases. Our system predicts protein content using Random Forest Regressor (MAE = 1.33g), classifies foods into Low/Medium/High Quality grades using Gradient Boosting (96.77% accuracy), and recommends foods and 200+ calculated Indian meals using Cosine Similarity with hard constraint filtering.'*

3. TOP 10 EXPECTED VIVA QUESTIONS & ANSWERS

Q1: What is the main objective and novel contribution of your project?

Most existing food nutrition ML prototypes depend exclusively on western databases (USDA), making them ineffective for Indian dietary patterns. Our contribution is integrating an Indian food composition dataset (1,014 dishes) and formulating a composite meal calculation framework (200 recipes) while achieving 96.77% classification accuracy and constraint-based recommendation.

Q2: Why did you use multiple data sources (USDA + Open Food Facts + Indian Data)?

Real-world data is heterogeneous. Different sources have different field names, measurement units, missingness, and geographic coverage. Integrating USDA, Open Food Facts, and Indian Food data creates a multi-source data engineering problem and proves our pipeline can handle schema normalization and deduplication.

Q3: Why did you decide NOT to blindly scrape tables from the ICMR-NIN IFCT 2017 PDF?

Scraping complex multi-page PDF publications causes header duplication, column misalignment, merged cells, missing values converted to zeroes, and OCR errors. To maintain scientific integrity and reproducibility, we used verifiable structured food datasets instead of raw PDF scraping.

Q4: How did you construct the 200 Composite Indian Meal dataset?

We defined representative meal combinations (e.g., Matar Paneer with Paneer Butter Sauce) with constituent ingredient weights (W_i in grams). We matched ingredient strings using token matching and calculated nutrition via: Meal_Nutrient = SUM [(W_i / 100) * Ingredient_Nutrient_100g].

Q5: How does your Content-Based Recommender work? Why Cosine Similarity over Collaborative Filtering?

Public nutrition databases lack user rating histories required for collaborative filtering. We used Content-Based Vector Cosine Similarity: Sim(A, B) = (A . B) / (||A|| ||B||) after standardizing features with StandardScaler. It ranks candidate foods against user target vectors.

Q6: How does hard constraint filtering and fallback logic work?

Before vector similarity ranking, candidates are filtered by numerical bounds (e.g. Max Calories <= 400, Min Protein >= 10g). If no food satisfies all constraints, the system automatically falls back to closest similarity matches with a clear warning so it never crashes.

Q7: How is the Nutritional Quality Score formulated? Is it a medical claim?

It is an educational scoring framework, NOT clinical advice. Core nutrients are normalized to [0, 1] using 1st-99th percentile scaling: Score = 0.35(Protein) + 0.25(Fiber) - 0.15(Sugar) - 0.10(Sodium) - 0.10(Fat) - 0.05(Calories). Scores (0-100) are binned into Low (<33.3), Medium (33.3-66.7), and High (>66.7) labels.

Q8: What do your Baseline vs. Expanded experiment results prove?

Random Forest Regressor achieved MAE = 1.33g on the expanded dataset vs 1.40g on baseline. Gradient Boosting Classifier achieved 96.77% accuracy on expanded data. This proves that adding Indian data preserves model performance while expanding regional recommendation utility.

Q9: How did you handle missing values and outliers?

Missing values were profiled globally, then imputed inside sklearn Pipelines using median imputation to prevent train/test data leakage. Outliers were analyzed using IQR box plots and retained if they represented genuine high-fat or high-protein food properties.

Q10: How did you test and verify your project?

We wrote automated unit tests in tests/test_pipeline.py testing meal calculation, standardization, quality label validity, ingredient matching, and constraint filtering (5/5 tests passed in Pytest).